Lab Legends

Instructor Setup & Classroom How-To

Windows + Docker Desktop | Local classroom network

What is Lab Legends?

Lab Legends is a browser-based, multiplayer classroom quiz. The instructor runs the game on one computer; students join from phones, tablets, or computers using a web browser. It is designed for a small class of up to 9 student players/teams on the same trusted local network.

The current package includes virtualization review questions and a Test Review round. It is a classroom prototype, so test it with a few devices before using it with a full class.

1. What you need

• A Windows computer to host the game.

• Docker Desktop installed and running. On Windows, Docker Desktop may require WSL 2 and a restart during setup.

• The Lab Legends updated ZIP package: lab_legends_change_answer.zip.

• Student devices connected to a network that allows devices to communicate with each other. Guest Wi-Fi or client isolation can block this.

• A browser on the instructor and student devices.

2. Start the game on the instructor computer

Download and extract lab_legends_change_answer.zip into a folder, for example C:\LabLegends.

Open the extracted folder. Confirm it contains docker-compose.yml.

In File Explorer, click the address bar, type powershell, and press Enter. This opens PowerShell in that folder. Alternatively, open a terminal in the folder.

Run the command below and wait for it to finish building and starting the app:

```
docker compose up -d --build
```

On the instructor computer, open a browser and go to:

```
http://localhost:3000
```

Choose “Launch instructor console” and start a room. Keep the instructor screen open; it displays the room PIN and the current question.

3. Let students join

On the instructor computer, open Command Prompt and run:

```
ipconfig
```

Find the active Wi-Fi or Ethernet adapter and note its IPv4 Address. Example only: 192.168.1.50. Use the actual address shown on your computer.

Give students this address with port 3000. Example only:

```
http://192.168.1.50:3000
```

Students open that address in a browser, enter the room PIN shown on the instructor screen, and choose a player/team name.

Do not give students http://localhost:3000. “localhost” means the device they are currently using, not the instructor computer.

4. Allow connections through Windows Firewall (if needed)

If students cannot open the page, Windows Firewall may be blocking incoming connections. On the instructor computer, open Windows Terminal or PowerShell using “Run as administrator.” Then run:

```
New-NetFirewallRule -DisplayName "Lab Legends TCP 3000" -Direction Inbound -Protocol TCP -LocalPort 3000 -Action A
```

A successful result shows Enabled: True and Status: OK. Run the command only once; if the rule already exists, you do not need to recreate it.

Use this rule only on a trusted classroom network. Check Settings > Network & internet > Wi-Fi (or Ethernet) and confirm the network profile is Private if that is appropriate for your institution’s network policy. Do not change institutional network settings without permission.

5. Run a question

• The instructor console shows the question and the same answer choices students see.

• Students select an answer and press “Lock In Answer.”

• Before the instructor reveals the answer, students can choose “Change Answer,” select a different option, and lock it in again.

• Only the final locked-in answer is scored.

• When ready, the instructor chooses “Reveal answer.” The correct choice is highlighted and vote counts are shown.

• Use the instructor controls to move to the next question and continue the game.

6. Troubleshooting

The game opens on the instructor computer but not on student devices

• Make sure students use the instructor computer’s IPv4 address and :3000, not localhost.

• Confirm all devices are on a network that permits device-to-device traffic. Guest Wi-Fi and wireless client isolation commonly block this.

• Check the Docker port mapping with the command below. Run it from the folder containing docker-compose.yml:

```
docker compose ps
```

Look for a mapping similar to 0.0.0.0:3000->3000/tcp. If the app is not running or no port mapping appears, check the output from Docker and start the app again.

Test whether port 3000 is reachable

From PowerShell on a student computer, replace the example IP with the instructor computer’s actual IPv4 address:

```
Test-NetConnection 192.168.1.50 -Port 3000
```

TcpTestSucceeded: True means the TCP connection succeeded. False means the service is not reachable; check Docker, the IP address, Windows Firewall, and network isolation.

Port 3000 is already in use

Stop the old version from its original folder, if it is still running:

```
docker compose down
```

Then start the desired version from the folder containing the updated docker-compose.yml:

```
docker compose up -d --build
```

The page still does not open

From the game folder, run docker compose ps and docker compose logs --tail=100. Save the output for your IT support person or administrator.

7. Stop the game

When finished, open a terminal in the folder containing docker-compose.yml and run:

```
docker compose down
```

This stops and removes the running Compose containers for this game. Start it again later with docker compose up -d.

Safety and usage notes

• Keep the game on a trusted local classroom network.

• Do not forward port 3000 on the router or expose this prototype directly to the public internet.

• Follow your institution’s policies for student names and classroom technology.

• Test the full flow with the instructor computer and two student devices before class. This package has not been verified in a live session with a full class.